



A/B Testing of Online Advertising



Which is Better ?



Agenda

- Problem Statement
- Visualizations and Analysis
- Recommendation and Insights

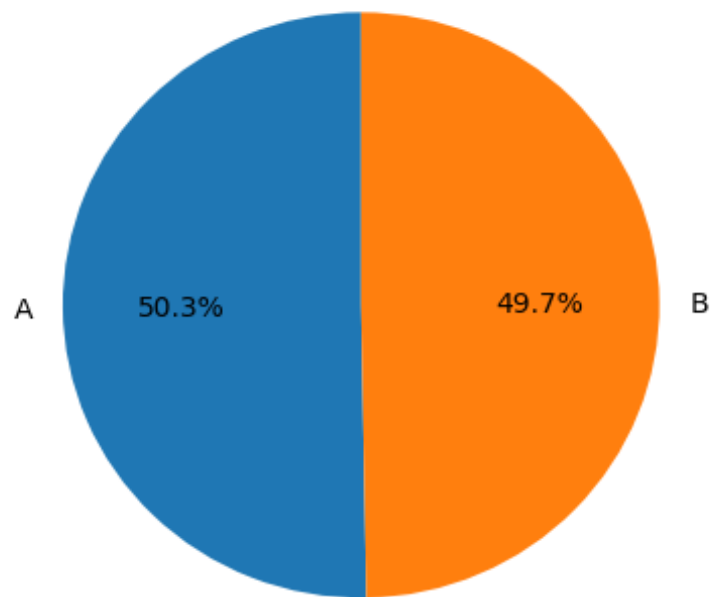


Problem Statement

The goal of this project is to analyze and compare the performance of two different product advertisements (Product A and Product B). The objective is to determine which product's advert leads to a higher click-through rate (CTR) and estimate the probability that one product performs better than the other.

Users By Product

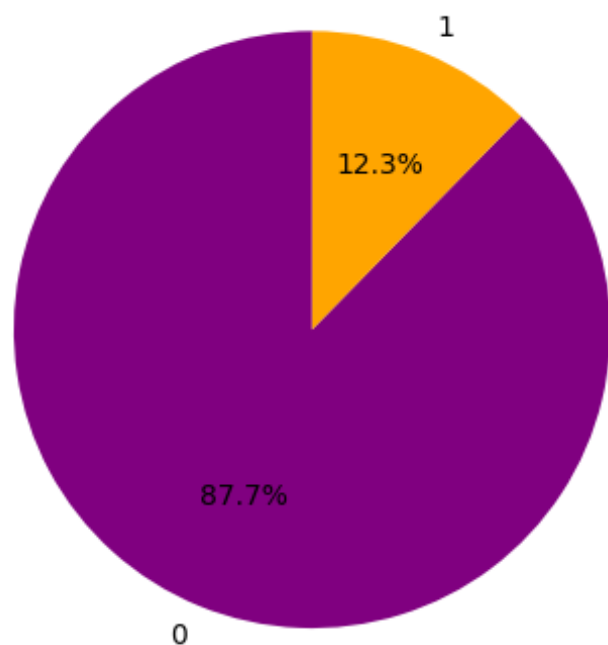
No of Users For Each Product



Product A has 50.3% users and Product B has 49.7% users. No of Users are close for both But maybe Product A is More engaging.

Users By Click Rate

Users Click Rate 1 = Yes , 0 = No



No of Users that clicked are 12.3% and Non-click users are 87.7% . Need to Identify Why users are not Clicking Need to Focus on Content



Hypothesis Testing

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Clicks A: 654 / 5028 | CTR: 0.1301  
Clicks B: 580 / 4972 | CTR: 0.1167  
Z-statistic: 2.0399  
P-value: 0.0414  
✅ Statistically significant difference between A and B ( $p < 0.05$ )
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Performed Z-Test to
Identify the Effectiveness
of product . Product A is
Better as Compared to
Product B

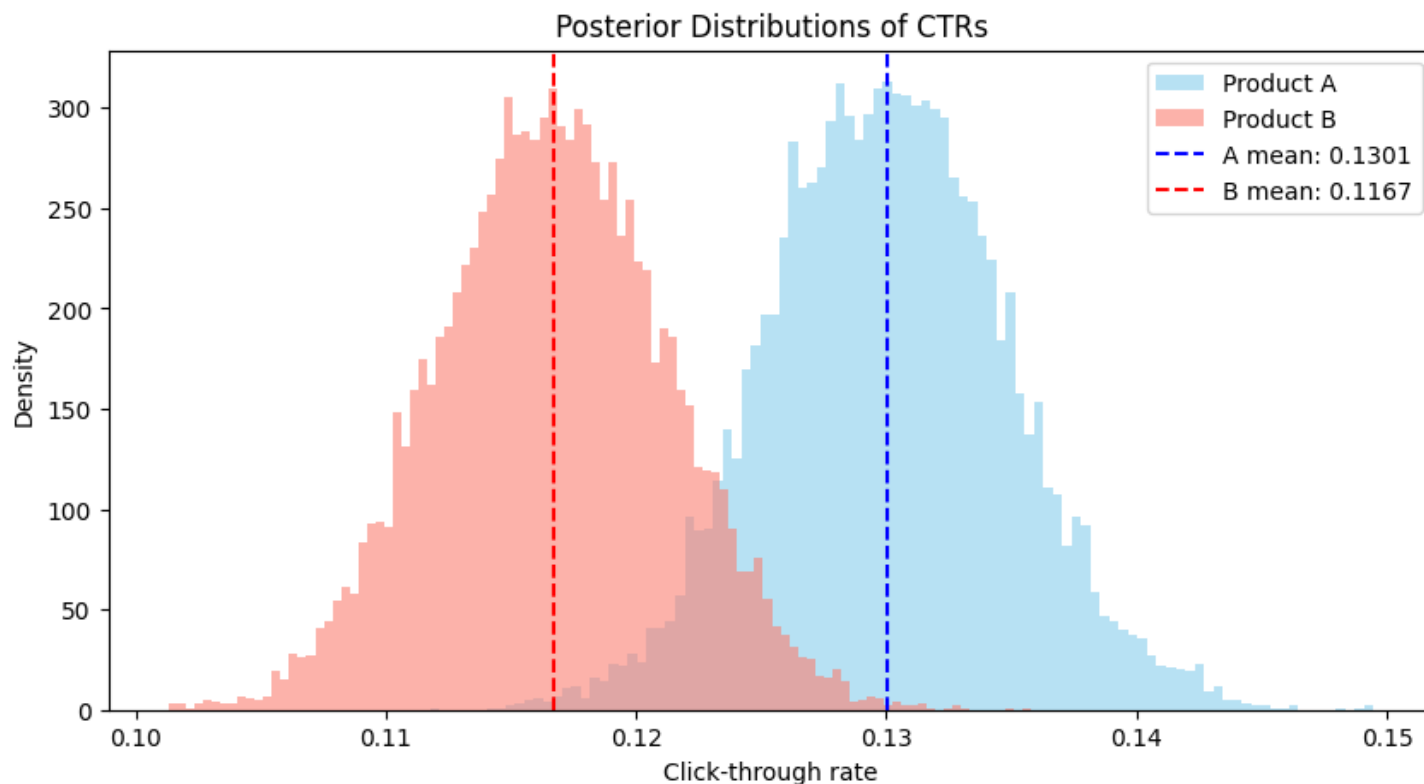


Bayesian A/B Testing

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Posterior A: Beta(655, 4379)  
Posterior B: Beta(581, 4397)  
Probability A > B: 0.9797
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Performed Bayesian Testing to Determine the Probability of Product A Effectiveness which is 97% more Accurate as compared to Product B

Conclusion



Product A is Better the Company should move with Product A . Also company should focus on marketing to increase the click rate .